

AP Physics C Electricity and Magnetism

APPENDIX I

FRQ Focus Lecture

Experiment-based FRQ Questions

“It’s all about graphing and linearization.”

Goal: determine the value of an unknown physics quantity or constant.

Measurements: record or measure two variables related to the unknown.

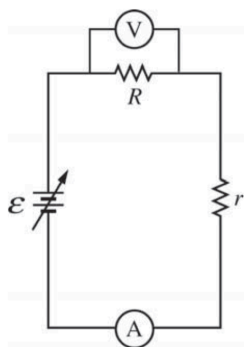
Linearization: find two quantities with linear relationship $y = k \cdot x + b$.

Graphing:

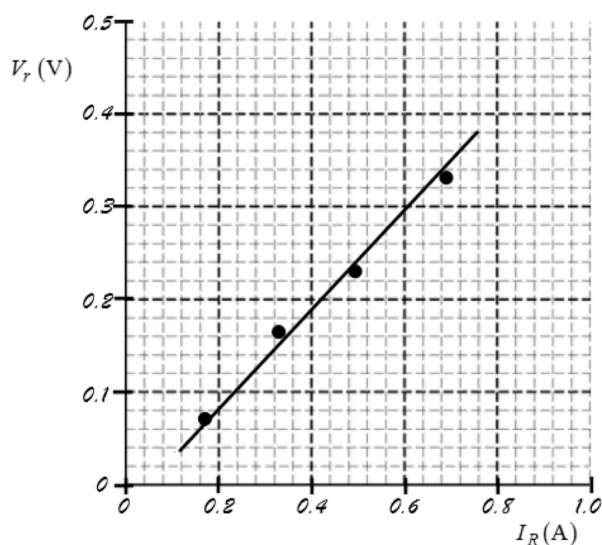
- 1) Set scale *Use more than half of each axis*
- 2) Label axes (with units!)
- 3) Plot points
- 4) Draw trendline (best-fit line)

Calculation:

- 5) Find slope *DO NOT USE DATA POINTS!*
- 6) Connect slope to “known” equations and solve for unknown value



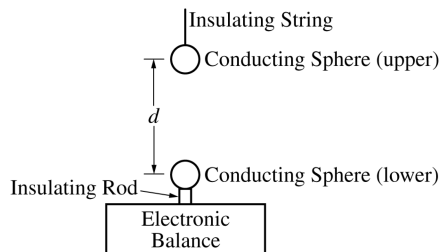
$\mathcal{E}$ (V)	V_R (V)	I_R (A)		
0.250	0.179	0.162		
0.500	0.335	0.327		
0.750	0.520	0.490		
1.000	0.670	0.687		





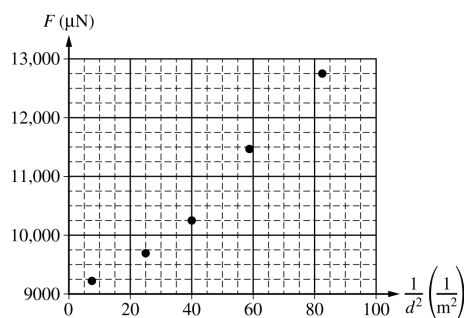
Unit 1: Electrostatics

1.(2021_NA_FRQ_2)



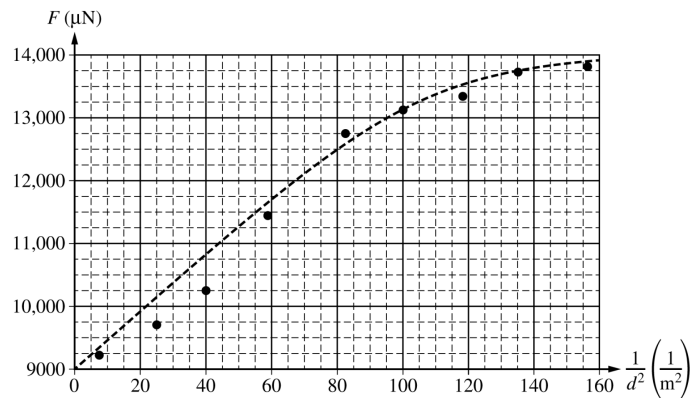
Students perform an experiment to study the force between two charged objects using the apparatus shown above, which contains two identical conducting spheres. The upper sphere is attached to an insulating string, which can be used to move the sphere downward. The lower sphere sits on an insulating rod, which is on an electronic balance. The electronic balance is zeroed before the lower sphere and insulating rod are in place.

For the first trial, a charge of Q is placed on each sphere and then the upper sphere is slowly moved downward. The students measure the distance d between the centers of the spheres and the magnitude F of the force that appears on the electronic balance. The recorded data are shown on the graph of F as a function of $\frac{1}{d^2}$ shown below.



- (a)
- Draw a line that represents the best fit to the points shown.
 - Use the graph to calculate the charge Q .
 - On the graph on the previous page, draw a circle around the data point that was taken when the distance between the centers of the spheres was the least.
 - Determine the distance between the centers of the spheres for the data point indicated above.
 - What physical quantity does the vertical intercept represent?

Justify your answer.



The experiment is extended by collecting additional data points, which appear on the right side of the graph shown above. The new data points do not follow the linear pattern seen with the first points. The group of students tries to explain this discrepancy.

(b) One student suspects that charge is slowly leaking off the top sphere. Could this explain the discrepancy?

_____ Yes _____ No

Justify your answer.

(c) A second student suspects that the excess charges have rearranged themselves, polarizing the spheres.

i. On the circles representing the spheres below, use a single “+” sign on each sphere to represent the locations of highest concentration of the excess positive charges.



ii. Explain how this rearrangement could be responsible for the discrepancy.

(d) A third student suggests that the experiment be modified so that the top sphere is given a negative charge that is equal in magnitude to the positive charge given to the bottom sphere.

i. On the circles representing the spheres below, use a single “+” sign on the bottom sphere to represent the location of highest concentration of the excess positive charges. Use a single “−” sign on the top sphere to represent the location of the highest concentration of the excess negative charges.



ii. For a separation distance equal to that of the data point indicated in part (a)(iii), would the magnitude of the force reading with spheres of opposite charges be greater than, less than, or equal to the magnitude of the force reading with spheres of the same charges?

_____ Greater than _____ Less than _____ Equal to

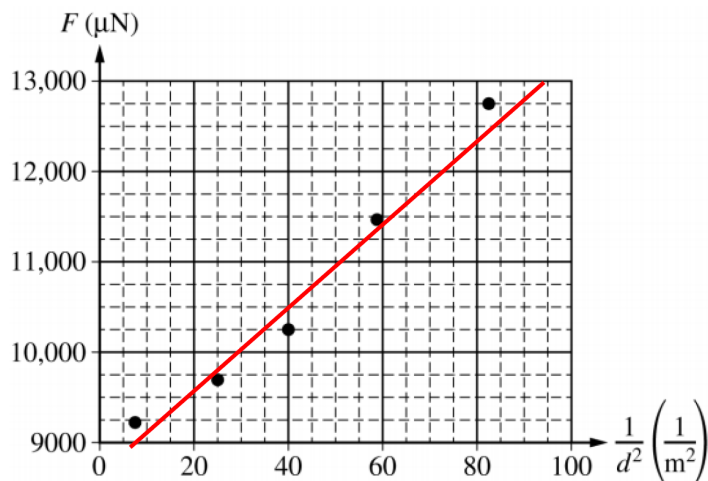
Justify your answer.

Scoring Guidelines

- (a) i. For drawing an appropriate best-fit line

1 point

Example response for part (a)(i)



- ii. For calculating the slope using two points from the best-fit line

1 point

$$\text{slope} = m = \frac{\Delta y}{\Delta x} = \frac{(12,000 - 10,500) \mu\text{N}}{(72 - 40) \text{ m}^{-2}} = 4.7 \times 10^{-5} \text{ N}\cdot\text{m}^2$$

For correctly relating the slope to the charge

1 point

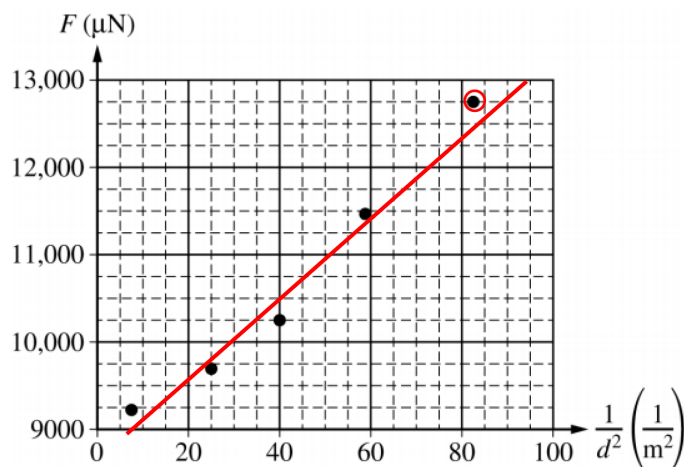
$$F = \frac{kQ^2}{r^2} \therefore \text{slope} = kQ^2$$

For correctly calculating the charge consistent with the slope from the best-fit line

1 point

$$Q = \sqrt{\frac{\text{slope}}{k}} = \sqrt{\frac{(4.7 \times 10^{-5} \text{ N}\cdot\text{m}^2)}{(9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)}} = 7.2 \times 10^{-8} \text{ C}$$

- (a) iii. For drawing a circle around the appropriate data point 1 point



- iv. For a correct answer 1 point

From the graph, $\frac{1}{d^2} = 82.5 \text{ m}^{-2}$

$$d = 0.11 \text{ m}$$

- v. For indicating that the y-intercept should be the weight of the conducting sphere and insulating rod 1 point

For a correct justification 1 point

Example responses for part (a)(v).

When the spheres are infinitely far apart, there is no electric force between them.

The only force recorded by the balance must be the weight of the lower sphere and insulating rod.

OR

The scale is zeroed before the insulating rod, and lower sphere are placed on it, so the nonzero y-intercept is their weight.

Total for part (a) 8 points

- (b) For selecting “Yes” and providing a correct justification 1 point

Example response for part (b)

Because the later force measurements are lower than the expected values on the best-fit line, then a decrease in charge could explain the discrepancy seen in the data.



Physics C: Electricity and Magnetism

(c) i.	For correctly indicating the position of the excess positive charges	1 point
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	For correctly indicating that the net charge is positive	1 point
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Example response for part (c)(i)



ii.	For a correct justification	1 point
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Example response for part (c)(ii)

On a conductor, excess charges are free to move and will repel each other. The excess charges in the sphere will rearrange themselves until they are as far apart from each other as possible; thus, the repelling force decreases.

Total for part (c)	3 points
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(d) i.	For appropriately indicating the position of the excess charges on both spheres	1 point
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ii.	For selecting “Less than” and attempting a relevant justification	1 point
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	For a correct justification	1 point
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Example response for part (d)

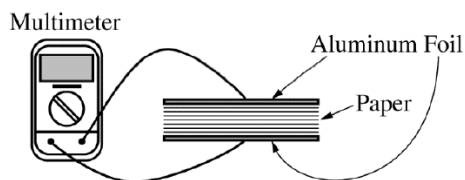
Because the opposite charged spheres will attract each other, this will decrease the upward normal force exerted by the scale on the lower sphere; thus, the reading on the scale will be less than when the spheres had the same charge.

Total for part (d)	3 points
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Total for question 2	15 points
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Unit 2: Capacitor and dielectrics

2.(2018_NA_FRQ_2)



An experiment is designed to measure the dielectric constant of paper that has an area $A = 0.060 \text{ m}^2$. Using aluminum foil, two parallel plates are created with the same area as the paper. Five hundred sheets of paper are placed between the aluminum foil plates to create a parallel plate capacitor, as shown in the figure above. Using a multimeter, the capacitance C of the capacitor is measured. The number of sheets and the total thickness d of the stack of paper are recorded. The experiment is repeated, reducing the number of sheets of paper each time. The data are recorded in the table below.

Sheets of Paper	d (m)	C (F)		
500	0.045	6.5×10^{-11}		
400	0.036	7.4×10^{-11}		
300	0.027	8.9×10^{-11}		
200	0.018	11.9×10^{-11}		
100	0.010	21.0×10^{-11}		

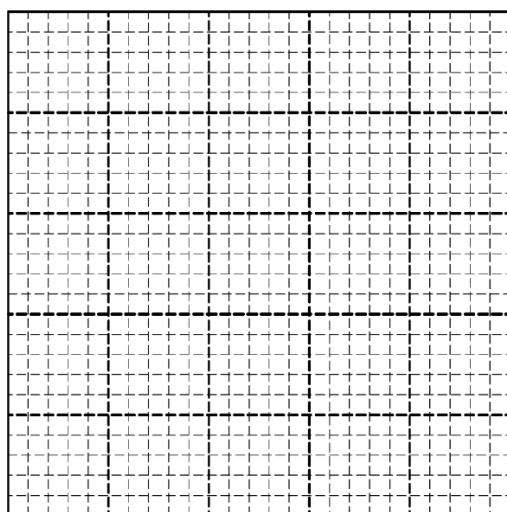
- (a) Indicate below which quantities should be graphed to yield a straight line whose slope could be used to calculate a numerical value for the dielectric constant of the paper.

Vertical axis: _____

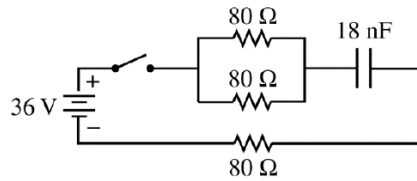
Horizontal axis: _____

Use the remaining columns in the table above, as needed, to record any quantities that you indicated that are not given. Label each column you use and include units.

- (b) Plot the data points for the quantities indicated in part (a) on the graph below. Clearly scale and label all axes, including units if appropriate. Draw a straight line that best represents the data.



- (c) Using the straight line, calculate a dielectric constant for the paper.



The student now makes a capacitor using the same aluminum foil plates and just one sheet of paper. Using the experimentally determined dielectric constant, the student calculates the capacitance to be $18\ \text{nF}$. The student uses this uncharged capacitor to build a circuit using wire, a $36\ \text{V}$ battery, 3 identical $80\ \Omega$ resistors, and an open switch, as shown in the figure above.

- (d) Calculate the current in the battery immediately after the switch is closed.
- (e) Determine the time constant for this circuit.
- (f) Students A and B measure the time it takes after the switch is closed for the voltage across the capacitor to reach half its maximum value and find that it is longer than expected.
- Student A assumes that the capacitance value is correct. Would Student A conclude that the resistance value is larger or smaller than measured?
 ____ Larger than measured ____ Smaller than measured
 Explain experimentally what could account for this.
 - Student B assumes that the resistance value is correct. Would Student B conclude that the capacitance value is larger or smaller than measured?
 ____ Larger than measured ____ Smaller than measured
 Explain experimentally what could account for this.



Scoring Guidelines

(a) 1 point

Indicate below which quantities should be graphed to yield a straight line whose slope could be used to calculate a numerical value for the dielectric constant of the paper.

Vertical axis: _____

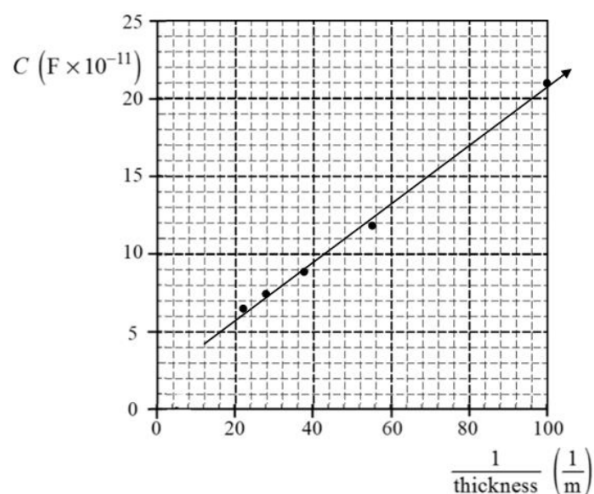
Horizontal axis: _____

Use the remaining columns in the table above, as needed, to record any quantities that you indicated that are not given. Label each column you use and include units.

For indicating variables that will create a straight line whose slope can be used to determine the dielectric constant of the paper		1 point
Example: Vertical axis: C Horizontal axis: $\frac{1}{d}$		
Note: Student earns full credit if the axes are reversed or if they use another acceptable combination.		

(b) 4 points

Plot the data points for the quantities indicated in part (a) on the graph below. Clearly scale and label all axes, including units if appropriate. Draw a straight line that best represents the data.



For a correct scale that uses more than half the grid		1 point
For correctly labeling the axes including units		1 point
For correctly plotting the data		1 point
For drawing a straight line consistent with the plotted data		1 point

(c) 3 points

Using the straight line, calculate a dielectric constant for the paper.

For correctly calculating the slope from the best-fit line and not the data points unless the points fall on the best-fit line	1 point
$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{(17 - 8)(\text{F} \times 10^{-11})}{(80 - 32)(1/\text{m})} = 0.19 \text{ F} \cdot \text{m}$ (Linear regression = 0.187 F•m)	
For correctly relating the slope to the dielectric constant	1 point
$\text{slope} = \kappa \epsilon_0 A$	
$\kappa = \frac{\text{slope}}{\epsilon_0 A} = \frac{(0.19 \text{ F} \cdot \text{m})}{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.060 \text{ m}^2)}$	
For a correct answer	1 point
$\kappa = 3.58$ (Linear regression = 3.52)	

(d) 3 points

Calculate the current in the battery immediately after the switch is closed.

For calculating the equivalent resistance for the parallel resistors	1 point
$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{80 \Omega} + \frac{1}{80 \Omega} = \frac{2}{80 \Omega} = \frac{1}{40 \Omega}$	
$R_p = 40 \Omega$	
For using Ohm's law with the potential difference across the capacitor equal to zero	1 point
$I = \frac{V}{R} = \frac{V}{R_p + R_3} = \frac{(36 \text{ V})}{R_p + R_3}$	
For substitution of values for resistance including the value for combined resistance above	1 point
$I = \frac{V}{R_p + R_3} = \frac{(36 \text{ V})}{(40 \Omega + 80 \Omega)} = 0.30 \text{ A}$	

(e) 2 points

Determine the time constant for this circuit.

For using the equation for the time constant with the equivalent resistance from above	1 point
$\tau = R_{eq} C = (120 \Omega)(18 \text{ nF})$	
For an answer with units consistent with part (d)	1 point
$t = 2.16 \mu\text{s}$	

(f) Students A and B measure the time it takes after the switch is closed for the voltage across the capacitor to reach half its maximum value and find that it is longer than expected.

i. 1 point

Student A assumes that the capacitance value is correct. Would Student A conclude that the resistance value is larger or smaller than measured?

____ Larger than measured ____ Smaller than measured

Explain experimentally what could account for this.

Select "Larger than measured"	
For an appropriate explanation	1 point
Example: The battery is not ideal and has internal resistance, so the actual resistance for the circuit is larger than the measured resistance.	



ii. 1 points

Student B assumes that the resistance value is correct. Would Student B conclude that the capacitance value is larger or smaller than measured?

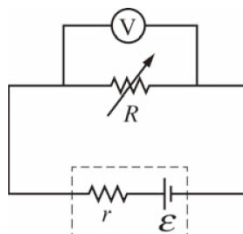
_____ Larger than measured _____ Smaller than measured

Explain experimentally what could account for this.

Select "Larger than measured"		
For an appropriate explanation		1 point
<i>Example: Some of the sheets of paper may be thinner than expected, so the actual capacitance for the circuit is larger than the measured capacitance.</i>		

Unit 3: Circuit

3.(2015_NA_FRQ_2)



E&M.2.

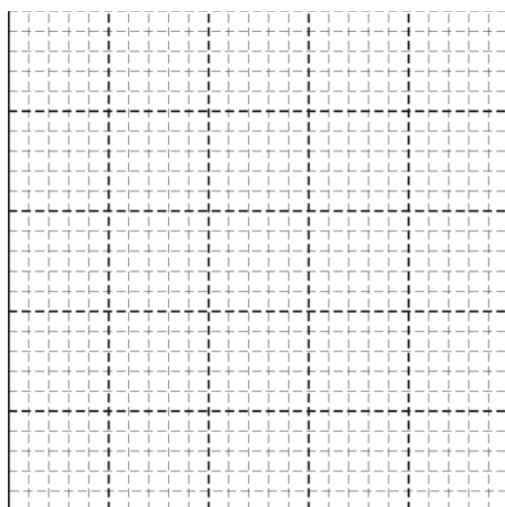
A student performs an experiment to determine the emf $\mathcal{E}$ and internal resistance r of a given battery. The student connects the battery in series to a variable resistance R , with a voltmeter across the variable resistor, as shown in the figure above, and measures the voltmeter reading V as a function of the resistance R . The data are shown in the table below.

Trial #	Resistance (Ω)	Voltage (V)	$1/R$ ($1/\Omega$)	$1/V$ ($1/V$)
1	0.50	5.6	2.00	0.179
2	1.0	7.4	1.00	0.135
3	2.0	9.4	0.50	0.106
4	3.0	10.6	0.33	0.094
5	5.0	10.9	0.20	0.092
6	10	11.4	0.10	0.088

(a)

- Derive an expression for the measured voltage V . Express your answer in terms of R , $\mathcal{E}$, r , and physical constants, as appropriate.
- Rewrite your expression from part (a)-i to express $1/V$ as a function of $1/R$.

(b) On the grid below, plot data points for the graph of $1/V$ as a function of $1/R$. Clearly scale and label all axes, including units as appropriate. Draw a straight line that best represents the data.



- (c) Use the straight line from part (b) to obtain values for the following.
- $\mathcal{E}$
 - r
- (d) Using the results of the experiment, calculate the maximum current that the battery can provide.
- (e) A voltmeter is to be used to determine the emf of the battery after removing the battery from the circuit. Two voltmeters are available to take this measurement—one with low internal resistance and one with high internal resistance. Indicate which voltmeter will provide the most accurate measurement.
- _____ The voltmeter with low resistance will provide the most accurate measurement.
- _____ The voltmeter with high resistance will provide the most accurate measurement.
- _____ The two voltmeters will provide equal accuracy.
- Justify your answer.

Scoring Guidelines

(a)

i. 2 points

Using Ohm's law:

$$V = IR$$

For a correct application of Kirchhoff's loop rule

$$\mathcal{E} = Ir + IR$$

$$I = \frac{\mathcal{E}}{(r + R)}$$

For a correct expression for the measured voltage across the variable resistor

$$V = \frac{\mathcal{E}}{(r + R)}R$$

1 point

1 point

ii. 1 point

For an expression of $1/V$ as a function of $1/R$ consistent with answer from part

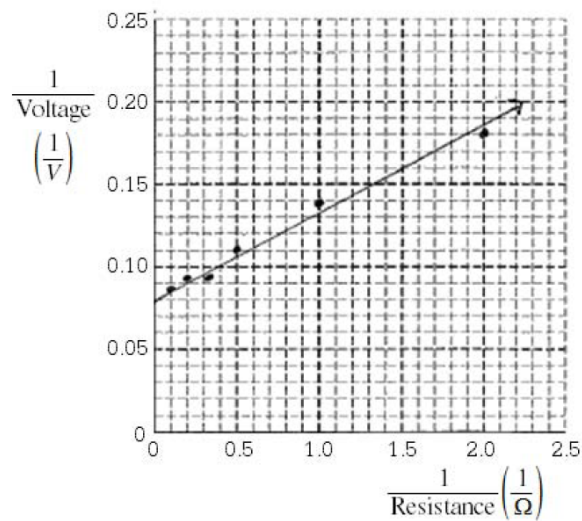
1 point

(a)(i)

$$\frac{1}{V} = \left(\frac{r}{\mathcal{E}}\right)\frac{1}{R} + \frac{1}{\mathcal{E}}$$

(b)

4 points



For correctly labeling both axes with variables and units

1 point

For correctly scaling both axes with an acceptable and appropriate scale

1 point

For correctly plotting the data points

1 point

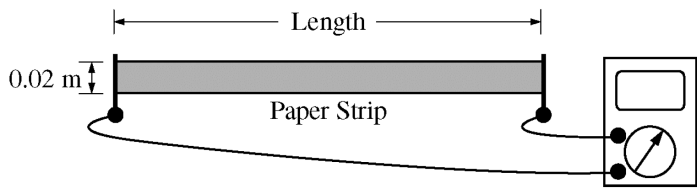
For correctly drawing a straight line that best represents the data

1 point

- (c)
- i. 2 points
- For using a value for the y -intercept consistent with the straight line drawn in part (b) 1 point
- $$y = mx + b$$
- $$b = 0.080 \text{ 1/V}$$
- For a correct substitution into the equation from part (a)(ii) 1 point
- $$\frac{1}{V} = \frac{(r + R)}{\mathcal{E}} \frac{1}{R}$$
- $$\frac{1}{V} = \frac{r}{\mathcal{E}} \frac{1}{R} + \frac{1}{\mathcal{E}}$$
- $$b = 1/\mathcal{E}$$
- $$\mathcal{E} = 1/(0.080 \text{ 1/V})$$
- $$\mathcal{E} = 12.5 \text{ V}$$
- ii. 2 points
- For calculating the slope using the straight line from part (b) and not data points 1 point
- $$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{(0.151 - 0.100)}{(1.50 - 0.40)} = 0.0463 \text{ } \Omega/\text{V}$$
- For a correct substitution into equation from part (a)(ii) 1 point
- $$m = \frac{r}{\mathcal{E}}$$
- $$r = m\mathcal{E}$$
- $$r = (0.0463 \text{ } \Omega/\text{V})(12.5 \text{ V})$$
- $$r = 0.58 \text{ } \Omega$$
- (d) 2 points
- For using Ohm's law 1 point
- $$\mathcal{E} = Ir$$
- For a correct substitution of values from part (c): 1 point
- $$(12.5 \text{ V}) = I(0.58 \text{ } \Omega)$$
- $$I = 21.6 \text{ A}$$
- (e) 2 points
- For selecting "The voltmeter with high resistance" 1 point
- For a correct justification 1 point
- Example: The voltmeter acts like a resistor in a circuit with the battery. It will measure the potential difference across its own internal resistance. The higher its internal resistance, the closer its potential difference will be to the emf of the battery.



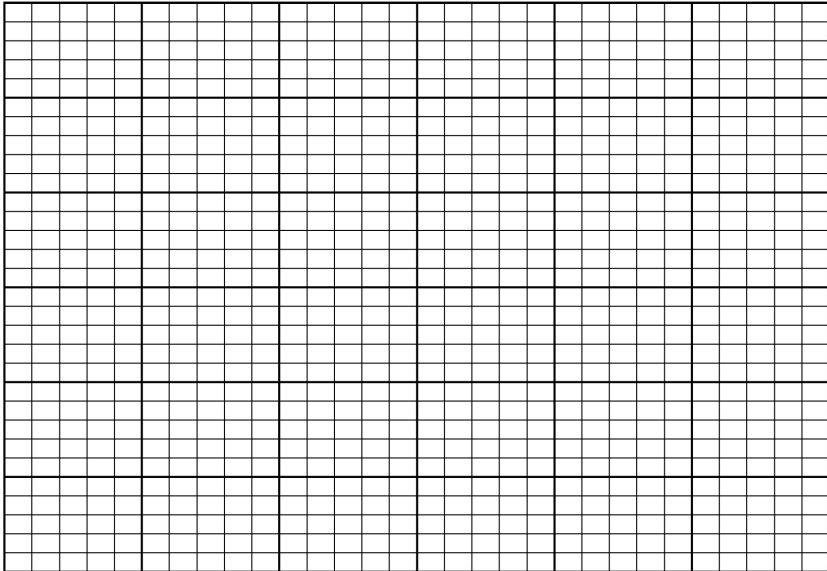
4.(2012_NA_FRQ_2)



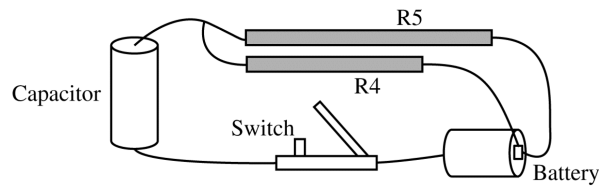
A physics student wishes to measure the resistivity of slightly conductive paper that has a thickness of 1.0×10^{-4} m . The student cuts a sheet of the conductive paper into strips of width 0.02 m and varying lengths, making five resistors labeled R1 to R5. Using an ohmmeter, the student measures the resistance of each strip, as shown above. The data are recorded below.

Resistor	R1	R2	R3	R4	R5
Length (m)	0.020	0.040	0.060	0.080	0.100
Resistance (Ω)	80,000	180,000	260,000	370,000	440,000

- (a) Use the grid below to plot a linear graph of the data points from which the resistivity of the paper can be determined. Include labels and scales for both axes. Draw the straight line that best represents the data.

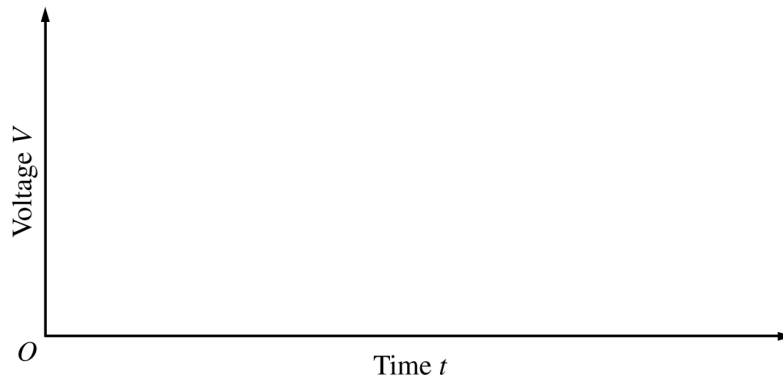


- (b) Using the graph, calculate the resistivity of the paper.



The student uses resistors R_4 and R_5 to build a circuit using wire, a 1.5 V battery, an uncharged $10\ \mu\text{F}$ capacitor, and an open switch, as shown above.

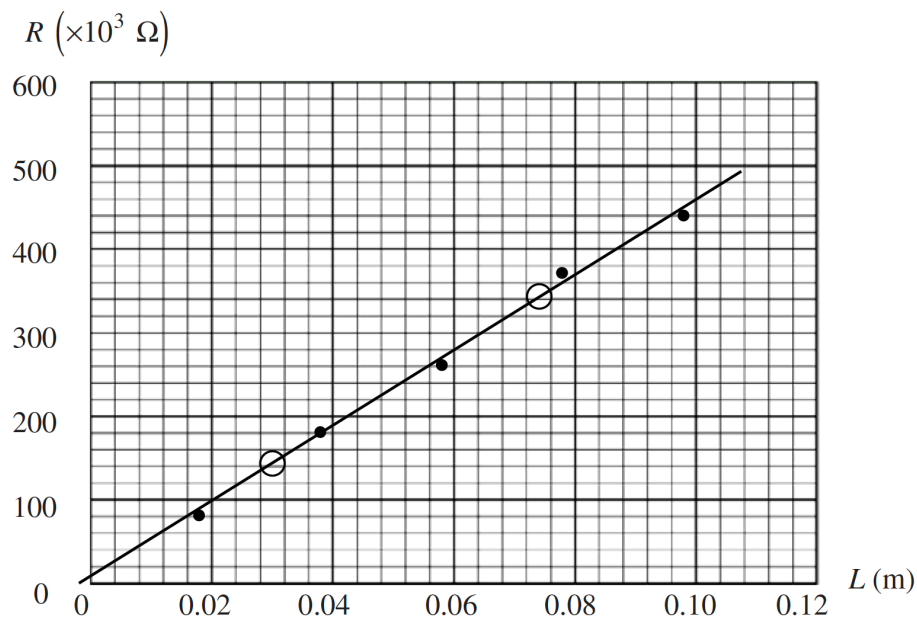
- (c) Calculate the time constant of the circuit.
- (d) At time $t = 0$, the student closes the switch. On the axes below, sketch the magnitude of the voltage V_c across the capacitor and the magnitudes of the voltages V_{R4} and V_{R5} across each resistor as functions of time t . Clearly label each curve according to the circuit element it represents. On the axes, explicitly label any intercepts, asymptotes, maxima, or minima with values or expressions, as appropriate.





Scoring Guidelines

(a) 4 points



For a correct label on each axis (one including resistance, one including length), that leads to a linear graph 1 point

For two linear scales, one for each axis, corresponding to the labels and occupying at least three-quarters of each axis 1 point

For reasonably correctly plotted points according to the scale 1 point

For a reasonable best-fit straight line 1 point

Note: Circles on the graph are to indicate points chosen to calculate slope in part (b). They are not necessary to receive credit.

(b) 3 points

For correctly using the equation $R = \rho L/A$ to solve for the resistivity

1 point

$$\rho = (R/L)A$$

For calculating the slope m from two points that lie on the indicated best-fit line

1 point

Example (using the two points indicated on the graph)

$$m = \Delta R / \Delta L = (340 \times 10^3 \, \Omega - 140 \times 10^3 \, \Omega) / (0.076 \, \text{m} - 0.032 \, \text{m})$$

$$m = 4.55 \times 10^6 \, \Omega/\text{m}$$

Note: The slope point can also be earned for indicating that a linear regression function on a calculator was used (exact answer required): $m = 4.5 \times 10^6 \, \Omega/\text{m}$ (to two significant figures).

$$\rho = mA = m\Delta L$$

$$\rho = (4.5 \times 10^6 \, \Omega/\text{m})(1.0 \times 10^{-4} \, \text{m})(0.02 \, \text{m})$$

For a correct answer, with supporting work

1 point

$$\rho = 9.0 \, \Omega \cdot \text{m} \, (\pm 0.2 \, \Omega \cdot \text{m} \text{ is acceptable})$$

(c) 3 points

For correctly stating the RC circuit time constant equation

1 point

$$\tau = RC$$

For the correct value of the equivalent resistance

1 point

$$\frac{1}{R_p} = \frac{1}{R_4} + \frac{1}{R_5} = \frac{1}{(370 \, \text{k}\Omega)} + \frac{1}{(440 \, \text{k}\Omega)}$$

$$R_p = 201 \, \text{k}\Omega$$

For the correct value of the time constant (units not evaluated here)

1 point

$$\tau = R_p C = (201 \, \text{k}\Omega)(10 \, \mu\text{F})$$

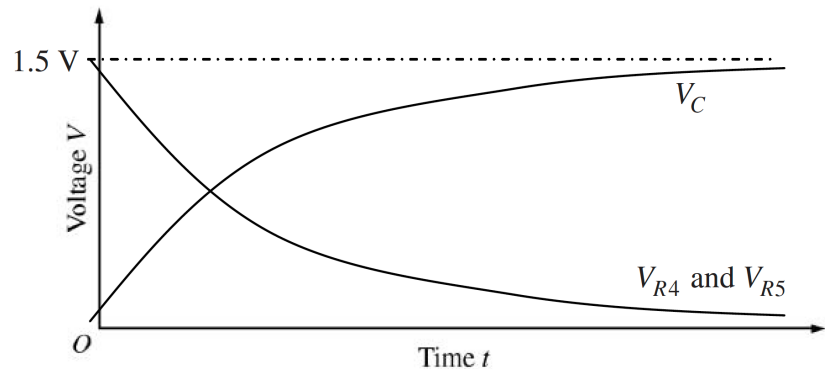
$$\tau = 2.01 \, \text{s}$$

Units 1 point

For correct units for the answers in both parts (b) and (c)

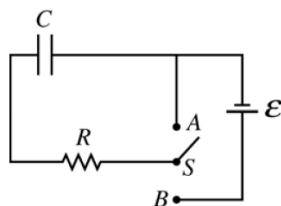
1 point

(d) 4 points



- | | |
|---|---------|
| For correctly indicating the battery voltage as an asymptote for the V_C curve (or in the absence of a correct V_C curve, as the maximum of the resistor curves) | 1 point |
| For a capacitor voltage curve starting at the origin, increasing, concave down, and appropriately labeled | 1 point |
| For a resistor voltage curve starting at the intersection of the asymptotic voltage with the voltage axis, decreasing and concave up (asymptotic with the time axis), and appropriately labeled | 1 point |
| For any indication that $V_{R4} = V_{R5}$ | 1 point |

5.(2013_NA_FRQ_2)



E&M 2.

In a lab, you set up a circuit that contains a capacitor C , a resistor R , a switch S , and a power supply, as shown in the diagram above. The capacitor is initially uncharged. The switch, which is initially open, can be moved to positions A or B .

(a)

- i. Indicate the position to which the switch should be moved to charge the capacitor.

___ A ___ B

- ii. On the diagram, draw a voltmeter that is properly connected to the circuit in a manner that will allow the voltage to be measured across the capacitor.

After a long time you move the switch to discharge the capacitor, and your lab partner starts a stopwatch. You collect the following measurements of the voltage across the capacitor at various times.

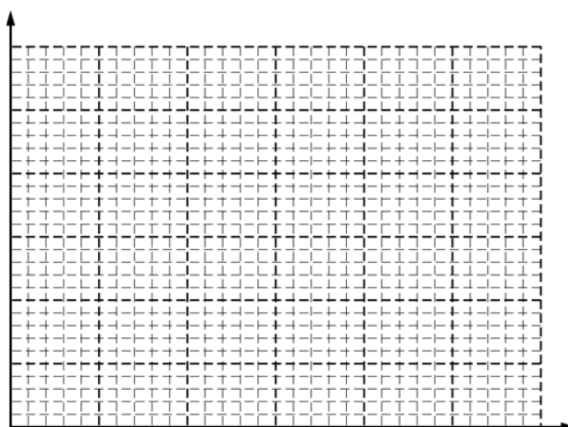
t (s)	6	18	30	42	54
V (V)	252	74	33	10	6

You wish to determine the time constant τ of the circuit from the slope of a linear graph.

(b)

- i. Indicate two quantities you would plot to obtain a linear graph.
- ii. Use the remaining rows in the table above, as needed, to record any quantities that you indicated that are not given. Label each row you use and include units.

- (c) On the axes below, graph the data from the table that will produce a linear relationship. Clearly scale and label all axes including units, if appropriate. Draw a straight line that best fits your data points.



- (d) From your line in part (c), obtain the value of the time constant τ of the circuit.



(e)

- i. In the experiment, the capacitor C had a capacitance of $1.50\ \mu\text{F}$. Calculate an experimental value for the resistance R .
- ii. On the axes in part (c), use a dashed line to sketch a possible graph if the capacitance was greater than $1.50\ \mu\text{F}$ but the resistance R was the same. Justify your answer.

Scoring Guidelines

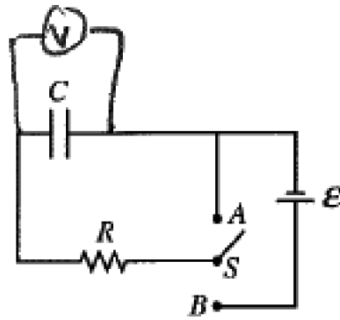
(a)

i. 1 point

For selecting B

1 point

ii. 1 point



For correctly drawing the voltmeter on the diagram in parallel with the capacitor

1 point

(b)

i. 2 points

For stating two correct variables that would yield a linear graph

2 points

Examples: t and $\ln V$ t and $-\ln V$ t and $\ln(V/V_0)$, where $V_0 = 252 \text{ V}$

Note: Students who do not come up with a correct pair of variables but who use Kirchhoff's loop rule to analyze the circuit, or write an appropriate differential equation or exponential relationship may earn one point.

ii. 1 point

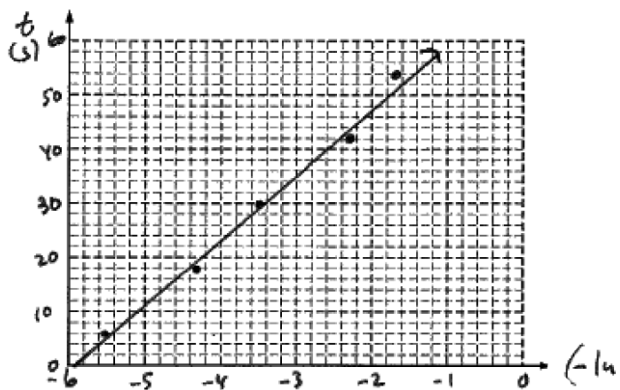
$t \text{ (s)}$	6	18	30	42	54
$V \text{ (V)}$	252	74	33	10	6
$-\ln V$	-6.53	-4.30	-3.50	-2.30	-1.79

For filling in values in a blank table row consistent with part (b) i

1 point

(c)

4 points



For correctly labeling the axes with variables and units consistent with part (b) 1 point
 For properly scaling the axes 1 point
 For correctly plotting data points consistent with part (b) 1 point
 For drawing a line of best fit through linear data 1 point

(d) 2 points

Relate the time constant to the slope from the graph above

$\tau = \text{slope}$ (or $1/\text{slope}$ if axes on graph are swapped)

For using points on the best fit line, not data points, to calculate the slope, or performing a linear regression 1 point

$$\tau = \frac{(52 - 23) \text{ s}}{-1.6 - (-4.0)}$$

For calculating a correct time constant 1 point

$$\tau = 12.1 \text{ s}$$

(e)

i. 2 points

For a correct expression relating the time constant to the resistance 1 point

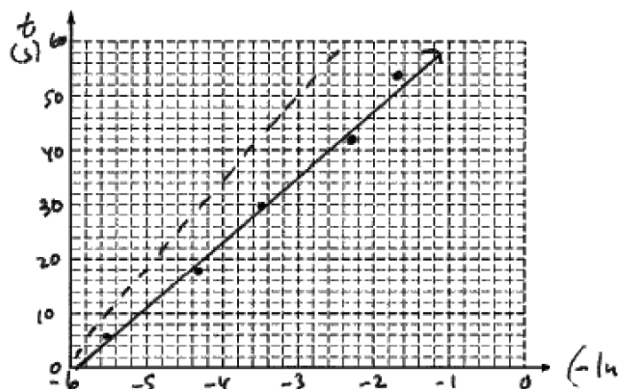
$$\tau = RC$$

For correctly substituting into the correct equation 1 point

$$R = \frac{\tau}{C} = \frac{(12.1 \text{ s})}{(1.5 \mu\text{F})}$$

$$R = 8.1 \text{ M}\Omega$$

ii. 2 points



For a dashed straight line with the same voltage at $t = 0$ as that of the solid line and a slope that is: 1 point

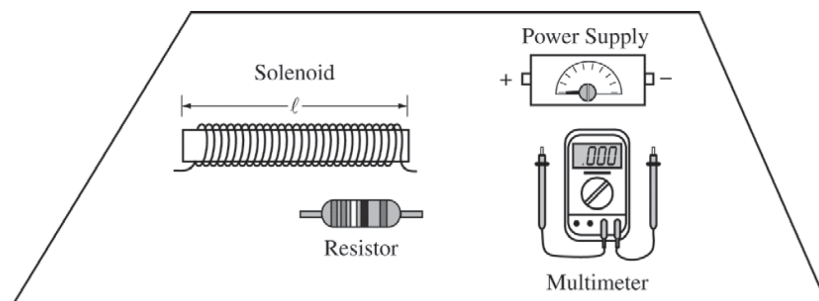
- *steeper* than that of the solid line, if t is graphed as a function of $\pm \ln V$, as shown above
- *shallower* than that of the solid line, if instead $\pm \ln V$ is graphed as a function of t

For a correct justification for the change in the slope that relates the slope to either the time constant or its reciprocal 1 point



Unit 4: Magnetic field

6.(2017_NA_FRQ_3)



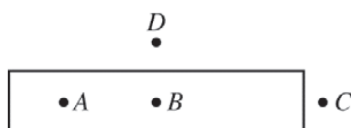
When studying Ampere’s law, students collect data on the magnetic field of two different solenoids in order to determine the magnetic permeability of free space μ_0 . The solenoids are created by wrapping wire around a hollow plastic tube. The solenoids of length ℓ with N turns of wire will be connected in series to a power supply and resistor. A multimeter will be used as an ammeter to measure the magnitude of the current I through the solenoids. The main components for the setup with one of the solenoids are shown in the figure above.

(a)

- On the figure above, draw wire connections between the solenoid, power supply, resistor, and multimeter that will complete the circuit and allow students to measure the magnitude of the current through the solenoid.
- Using the connections you made in part (a)i above, what will be the direction of the magnetic field inside the solenoid?

☐ Toward the top of the page ☐ To the left ☐ Out of the page
☐ Toward the bottom of the page ☐ To the right ☐ Into the page

The rectangle shown below represents the solenoid (the loops of wire are not shown). Points A , B , and C are along the central axis of the solenoid with point B at the middle of the solenoid. Point D is directly above point B .

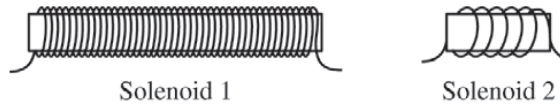


- From the choices below, select the point where you would place a magnetic field probe (a probe that can measure the magnitude of the magnetic field) to best measure the strength of the magnetic field of the solenoid in order to determine the magnetic permeability of free space μ_0 .

☐ A ☐ B ☐ C ☐ D

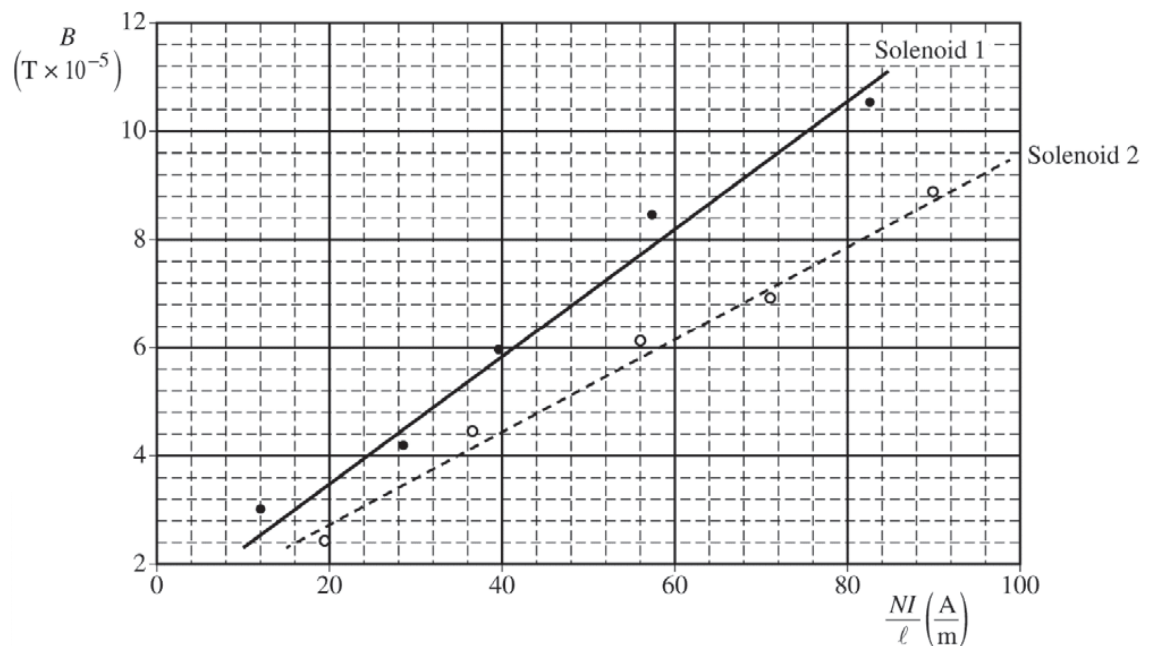
Justify your answer based on the model for a simple solenoid.

The figures below show two different solenoids that will be connected in the circuit above. Solenoid 1 has a length $\ell = 25$ cm with $N = 100$ turns. Solenoid 2 has a length $\ell = 5.0$ cm with $N = 5$ turns.



Note: Figures not drawn to scale.

A graph of the magnitude of the magnetic field B as a function of NI/ℓ is shown below. The best-fit lines for the data are shown as a solid line for solenoid 1 and as a dashed line for solenoid 2.



- (b) Which solenoid's best-fit line would give the best results for determining a value for the magnetic permeability of free space μ_0 ?

_____ Solenoid 1 _____ Solenoid 2

Justify your answer.

(c)

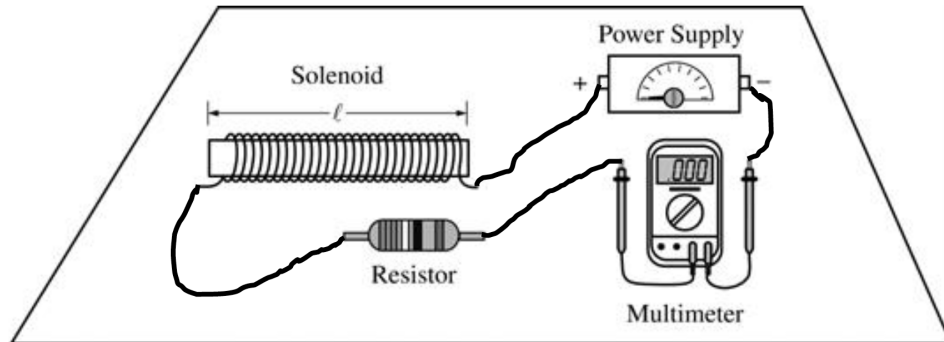
- Use the slope of the best-fit line for the solenoid chosen in part (b) to calculate the magnetic permeability of free space μ_0 .
- Calculate the percent error for the experimental value of the magnetic permeability of free space μ_0 determined in part (c)i.

(d)

- What is a reasonable physical explanation for a best-fit line that does not pass through the origin?
- Suppose a student connects the solenoid in a closed circuit similar to the circuit in part (a)i but without the resistor. The student notices the multimeter stops functioning after the power supply is turned on. Explain what causes the failure of the multimeter.

Scoring Guidelines

- (a)
i. 2 points



For connections that create a circuit with the power supply, solenoid, and resistor in series with each other 1 point

For connections that create a circuit with the multimeter in series with the inductor/resistor combination 1 point

- ii. 1 point

For selecting the answer consistent with the answer in part (a)(i) 1 point

- iii. 3 points

For selecting point *B* 1 point

For any indication that the magnetic field outside the solenoid is less than inside the solenoid 1 point

For any indication that the closer you are to the center of the solenoid, the better the approximation of an ideal solenoid due to the lack of edge effects 1 point

Example: The equation for the magnetic field for a solenoid is for an ideal solenoid. This does not work outside the solenoid (this eliminates *C* and *D*). In addition, as you approach the edge of the solenoid, the ideal approximation is not as accurate (this eliminates *A*). Thus, point *B* gives the best approximation of an ideal solenoid.

- (b) 2 points

For any indication that Solenoid 1 has more turns per unit length 1 point

For any indication that Solenoid 1 is longer and therefore more like an ideal solenoid 1 point

Examples:

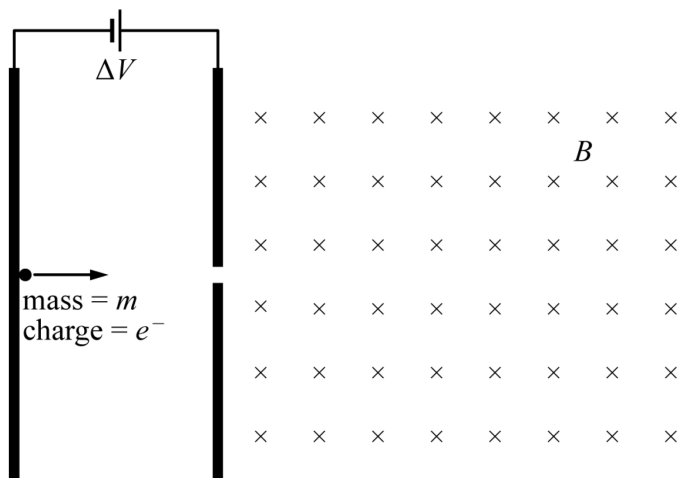
The equation for the magnetic field of a solenoid ($B = \mu_0 nI$) is based on an ideal solenoid in which you can ignore edge effects. Because Solenoid 1 has more turns per unit length, it is a better approximation of an ideal solenoid and thus its slope will give a better value for the permeability of free space.

The slope for Solenoid 1 is closer to the accepted value than the slope for Solenoid 2; therefore, the graph for Solenoid 1 is better.

- (c)
- i. 3 points
- For correctly calculating the slope for the solenoid chosen in part (b) using the best-fit line and not the data points 1 point
- $$\text{slope} = \frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{(10 - 4)(\text{T} \times 10^{-5})}{(75 - 24)(\text{A/m})} = 1.17 \times 10^{-6} \text{ T} \cdot \text{m/A}$$
- For correctly relating the magnetic permeability of free space to the slope 1 point
- $$\mu_0 = \text{slope}$$
- For correct units 1 point
- $$\mu_0 = 1.17 \times 10^{-6} \text{ T} \cdot \text{m/A}$$
- For Solenoid 2
- $$\text{slope} = \frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{(8 - 4)(\text{T} \times 10^{-5})}{(82 - 35)(\text{A/m})} = 8.51 \times 10^{-7} \text{ T} \cdot \text{m/A}$$
- ii. 1 point
- For using a correct equation for % error including the substitution of the accepted and the measured values 1 point
- $$\% \text{ error} = \frac{|\text{acc} - \text{exp}|}{\text{acc}} \times 100\% = \frac{|(1.26 \times 10^{-6}) - (1.18 \times 10^{-6})|}{(1.26 \times 10^{-6})} \times 100\% = 6.35\%$$
- For Solenoid 2
- $$\% \text{ error} = \frac{|\text{acc} - \text{exp}|}{\text{acc}} \times 100\% = \frac{|(1.26 \times 10^{-6}) - (8.51 \times 10^{-7})|}{(1.26 \times 10^{-6})} \times 100\% = 32.4\%$$
- (d)
- i. 1 point
- For a correct explanation involving any extraneous B fields 1 point
- Example: A component of Earth's magnetic field is in the direction of the inductor's magnetic field.
- ii. 2 points
- For stating that either the solenoid or the circuit has near zero resistance 1 point
- For stating that as a result of near zero resistance the current becomes very high 1 point
- Example: The inductor has near zero resistance. Therefore, without the resistor in series with it, the inductor will draw a near infinite current that the multimeter cannot handle.



7.(2019_NA_Set2_FRQ_3)



Two plates are set up with a potential difference V between them. A small sphere of mass m and charge $-e$ is placed at the left-hand plate, which has a negative charge, and is allowed to accelerate across the space between the plates and pass through a small opening. After passing through the small opening, the sphere enters a region in which there is a uniform magnetic field of magnitude B directed into the page, as shown above. Ignore gravitational effects. Express all algebraic answers in terms of V , m , e , B , and fundamental constants, as appropriate.

- (a)
- What is the initial direction of the force on the sphere as it enters the magnetic field?
 ____ Into the page ____ Out of the page
 ____ Toward the top of the page ____ Toward the bottom of the page
 - Describe the path taken by the sphere after it enters the magnetic field.
- (b) Derive an expression for the speed of the sphere as it passes through the small opening.
- (c) Derive an expression for the radius of the path taken by the sphere as it moves through the magnetic field.



Physics C: Electricity and Magnetism

An experiment is performed in which a beam of electrons is accelerated across the space between the plates and passes through the small opening. After passing through the opening, the electrons travel in a semicircular path and strike the right-hand plate. The potential difference between the plates is varied in regular increments, as shown in the table below. For each potential difference, the magnetic field is varied in order to cause the beam to strike the right-hand plate at a distance of 0.020 m from the opening.

Potential difference (V)	60	70	100	110	120	140
Magnetic field ($T \times 10^{-3}$)	2.62	2.78	3.39	3.54	3.78	3.99

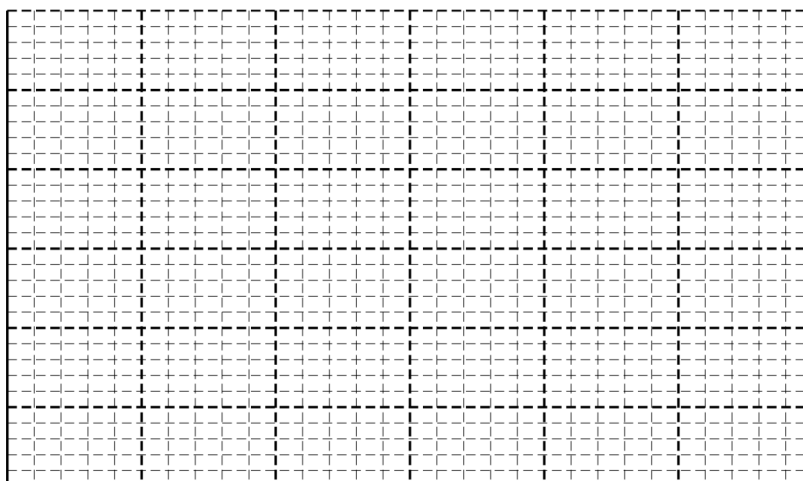
- (d) Indicate below which quantities should be graphed to yield a straight line whose slope could be used to calculate a numerical value for the mass-to-charge ratio of an electron.

Vertical axis: _____

Horizontal axis: _____

Use the remaining columns in the table above, as needed, to record any quantities that you indicated that are not given. Label each column you use and include units.

- (e) On the graph below, plot the relationship determined in part (d). Clearly scale and label all axes, including units, if appropriate. Draw a straight line that best represents the data.

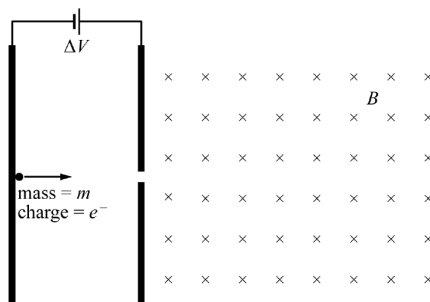


- (f) Using the straight line from part (e), determine the mass-to-charge ratio of an electron.



Scoring Guidelines

15 points



Two plates are set up with a potential difference V between them. A small sphere of mass m and charge $-e$ is placed at the left-hand plate, which has a negative charge, and is allowed to accelerate across the space between the plates and pass through a small opening. After passing through the small opening, the sphere enters a region in which there is a uniform magnetic field of magnitude B directed into the page, as shown above. Ignore gravitational effects. Express all algebraic answers in terms of V , m , e , B , and fundamental constants, as appropriate.

(a)

- i. LO CHG-1.A.a, SP 7.A
1 point

What is the initial direction of the force on the sphere as it enters the magnetic field?

☐ Into the page ☐ Out of the page
☐ Toward the top of the page ☐ Toward the bottom of the page

For selecting “Toward the bottom of the page”	1 point
---	---------

- ii. LO CHG-1.B, SP 7.A
1 point

Describe the path taken by the sphere after it enters the magnetic field.

For describing a circular path for the sphere consistent with the selection from part (a)(i)	1 point
Example: The sphere will move in a circular path toward the bottom of the page.	

- (b) LO CNV-1.E, SP 5.A, 5.E
2 points

Derive an expression for the speed of the sphere as it passes through the small opening.

For using a valid equation relating potential difference to kinetic energy of the sphere	1 point
$\Delta K = -\Delta U \therefore \frac{1}{2}mv^2 = -q\Delta V$	
$\frac{1}{2}mv^2 = eV$	
For the correct answer with supporting work	1 point
$v = \sqrt{\frac{2eV}{m}}$	

- (c) LO CHG-1.C, SP 5.A, 5.E
3 points

Derive an expression for the radius of the path taken by the sphere as it moves through the magnetic field.

For an expression relating the magnetic force to the centripetal force		1 point
$\frac{mv^2}{r} = Bqv$		
For substituting the charge of the sphere into the above equation and solving for r		1 point
$\frac{mv}{r} = Be \therefore r = \frac{mv}{Be}$		
For substituting the answer from part (b)		1 point
$r = \frac{m}{Be} \sqrt{\frac{2eV}{m}} = \sqrt{\frac{2mV}{eB^2}}$		

An experiment is performed in which a beam of electrons is accelerated across the space between the plates and passes through the small opening. After passing through the opening, the electrons travel in a semicircular path and strike the right-hand plate. The potential difference between the plates is varied in regular increments, as shown in the table below. For each potential difference, the magnetic field is varied in order to cause the beam to strike the right-hand plate at a distance of 0.020 m from the opening.

Potential difference (V)	60	70	100	110	120	140
Magnetic field ($T \times 10^{-3}$)	2.62	2.78	3.39	3.54	3.78	3.99

- (d) LO CHG-1.C, SP 4.C
1 point

Indicate below which quantities should be graphed to yield a straight line whose slope could be used to calculate a numerical value for the mass-to-charge ratio of an electron.

Vertical axis: _____
Horizontal axis: _____

Use the remaining columns in the table above, as needed, to record any quantities that you indicated that are not given. Label each column you use and include units.

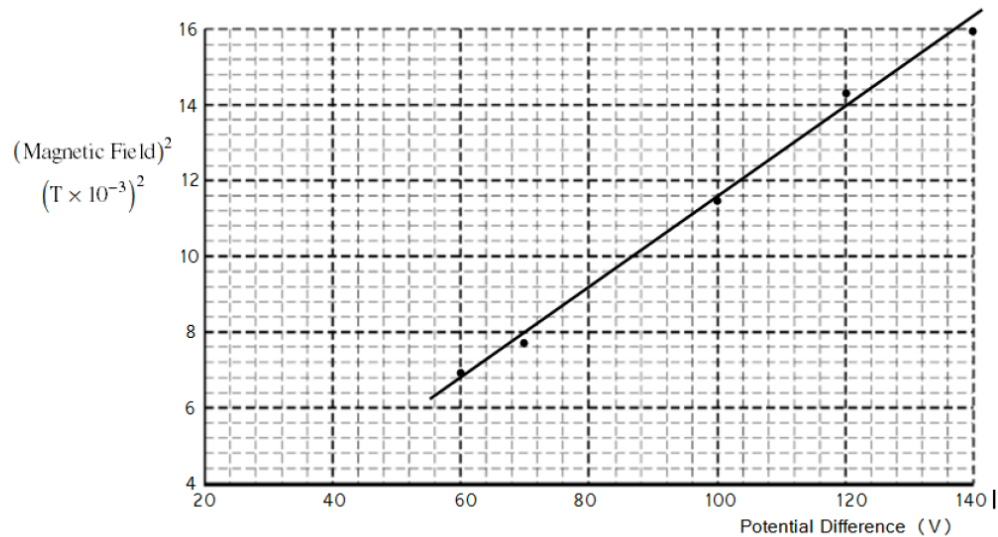
$$r = \sqrt{\frac{2m_e V}{eB^2}} \therefore B^2 = \frac{2m_e}{er^2} V$$

Potential difference (V)	60	70	100	110	120	140
Magnetic field ($T \times 10^{-3}$)	2.62	2.78	3.39	3.54	3.78	3.99
(Magnetic field) ² ($T \times 10^{-3}$) ²	6.86	7.73	11.5	12.5	14.3	15.9

For indicating axis choices with appropriate quantities to produce a straight line		1 point
Note: There are several possible choices.		

- (e) LO CHG-1.C, SP 3.A, 4.C
4 points

On the graph below, plot the relationship determined in part (d). Clearly scale and label all axes, including units, if appropriate. Draw a straight line that best represents the data.



For correctly labeling both axes with quantities and units	1 point
For correctly scaling and labeling the axes so that the points use at least half the grid	1 point
For correctly plotting the data	1 point
For drawing a straight line that represents the data	1 point

- (f) LO CHG-1.C, SP 4.D, 5.E
3 points

Using the straight line from part (e), determine the mass-to-charge ratio of an electron.

For correctly calculating the slope from the best-fit straight line and not from the data points unless the points fall on the best-fit line	1 point
$\text{slope} = \frac{14 - 9.2}{120 - 80} = 0.12 \frac{(\text{T} \times 10^{-3})^2}{\text{V}} = 1.2 \times 10^{-7} \text{ T}^2/\text{V}$	
For a correct expression relating the slope to the mass to charge ratio of an electron	1 point
$B^2 = \frac{2m_e}{er^2} V \therefore \text{slope} = \frac{2m_e}{er^2} \therefore \frac{m_e}{e} = \frac{r^2 \times \text{slope}}{2}$	
For substituting correct values into the above equation	1 point
$\frac{m_e}{e} = \frac{(0.01 \text{ m})^2 \times (1.2 \times 10^{-7} \text{ T}^2/\text{V})}{2} = 6.0 \times 10^{-12} \text{ kg/C}$	

Learning Objectives

CHG-1.A.a: Calculate the magnitude and direction of the magnetic force of interaction between a moving charged particle of specified charge and velocity moving in a region of a uniform magnetic field.

CHG-1.B: Describe the path of different moving charged particles (i.e., of different type of charge or mass) in a uniform magnetic field.

CHG-1.C: Derive an expression for the radius of a circular path for a charged particle of specified characteristics moving in a specified magnetic field.

CNV-1.E: Calculate the work done or changes in kinetic energy (or changes in speed) of a charged particle when it is moved through some known potential difference.

Science Practices

3.A: Select and plot appropriate data.

4.C: Linearize data and/or determine a best-fit line or curve.

4.D: Select relevant features of a graph to describe a physical situation or solve problems.

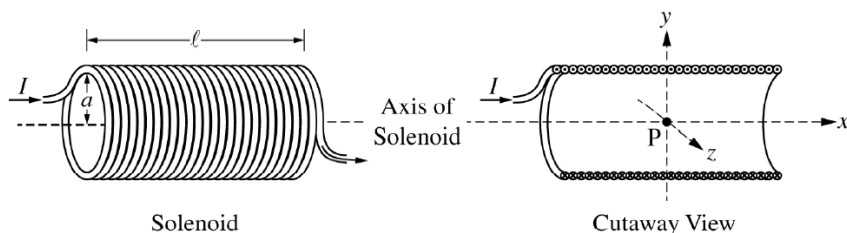
5.A: Select an appropriate law, definition, or mathematical relationship or model to describe a physical situation.

5.E: Derive a symbolic expression from known quantities by selecting and following a logical algebraic pathway.

7.A: Make a scientific claim.



8.(2019_NA_Set1_FRQ_3)



Note: Figures not drawn to scale.

A solenoid is used to generate a magnetic field. The solenoid has an inner radius a , length ℓ , and N total turns of wire. A power supply, not shown, is connected to the solenoid and generates current I , as shown in the figure on the left above. The x -axis runs along the axis of the solenoid. Point P is in the middle of the solenoid at the origin of the xyz -coordinate system, as shown in the cutaway view on the right above. Assume $\ell \gg a$.

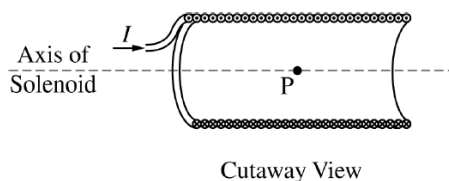
(a) Select the correct direction of the magnetic field at point P.

- ☐ $+x$ -direction ☐ $+y$ -direction ☐ $+z$ -direction
☐ $-x$ -direction ☐ $-y$ -direction ☐ $-z$ -direction

Justify your selection.

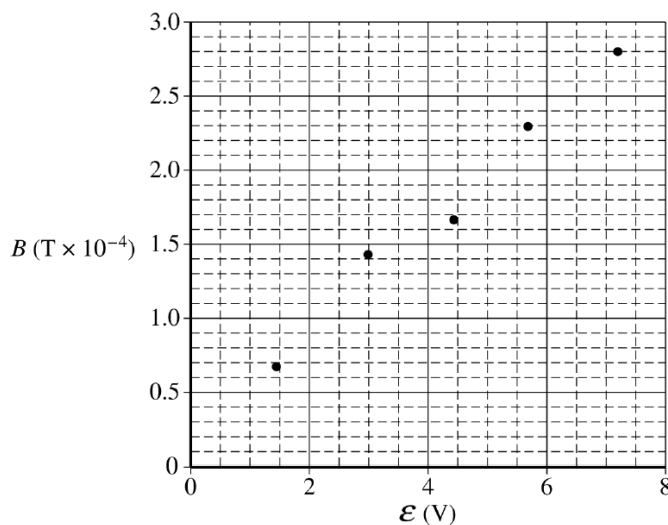
(b)

- i. On the cutaway view below, clearly draw an Amperian loop that can be used to determine the magnetic field at point P at the center of the solenoid.



- ii. Use Ampere's law to derive an expression for the magnetic field strength at point P. Express your answer in terms of I , ℓ , N , a , and physical constants, as appropriate.

Some physics students conduct an experiment to determine the resistance R_S of a solenoid with radius $a = 0.015$ m, total turns $N = 100$, and total length $\ell = 0.40$ m. The students connect the solenoid to a variable power supply. A magnetic field sensor is used to measure the magnetic field strength along the central axis at the center of the solenoid. The plot of the magnetic field strength B as a function of the emf $\mathcal{E}$ of the power supply is shown below.



(c)

- On the graph above, draw a best-fit line for the data.
- Use the straight line to determine the resistance R_S of the solenoid used in the experiment.

(d) One of the students notes that the horizontal component of the magnetic field of Earth is $2.5 \times 10^{-5} \text{ T}$.

- Is there evidence from the graph that the horizontal orientation of the solenoid affects the measured values for B ?

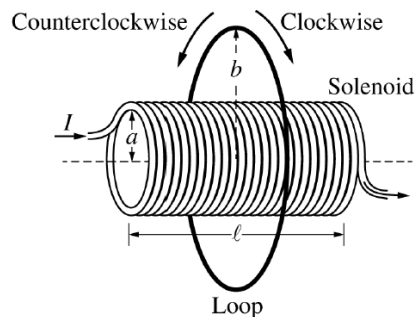
____ Yes ____ No

Justify your answer.

- Would the horizontal orientation of the solenoid affect the calculated value for R_S ?

____ Yes ____ No

Justify your answer.



A thin conducting loop of radius b and resistance R_L is placed concentric with the solenoid, as shown above. The current in the solenoid is decreased from I to zero over time Δt .

(e)

- Is the direction of the induced current in the loop clockwise or counterclockwise during the time period that the current in the solenoid is decreasing?

____ Clockwise ____ Counterclockwise

Justify your answer.

- Derive an equation for the average induced current i_{IND} in the loop during the time period that the current in the solenoid is decreasing. Express your answer in terms of I , ℓ , N , a , b , R_L , R_S , Δt , and physical constants, as appropriate.

Scoring Guidelines

- (a) LO CNV-8.E.a, SP 7.A, 7.C
2 points

Select the correct direction of the magnetic field at point P.

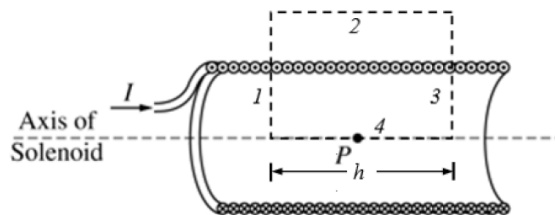
☐ +x-direction ☐ +y-direction ☐ +z-direction
☐ -x-direction ☐ -y-direction ☐ -z-direction

Justify your selection.

For choosing the “+x-direction” and providing a justification	1 point
For a correct justification	1 point
Example: Using the right-hand rule for current on the left side of the solenoid, the fingers curl into the loop, so the magnetic field points to the right, or in the +x-direction.	
Example: Using the right-hand rule for solenoids, when the fingers curl around the solenoid in the direction of the current, the thumb points to the right, therefore the magnetic field is to the right, or in the +x-direction.	

- (b)
i. LO CNV-8.C.c, SP 3.D
1 point

On the cutaway view below, clearly draw an Amperian loop that can be used to determine the magnetic field at point P at the center of the solenoid.



For drawing a rectangle with one side along the central axis of the solenoid and another side outside the solenoid and whose edges do not extend beyond the solenoid	1 point
--	---------

- ii. LO CNV-8.C.c, SP 5.A, 5.E
2 points

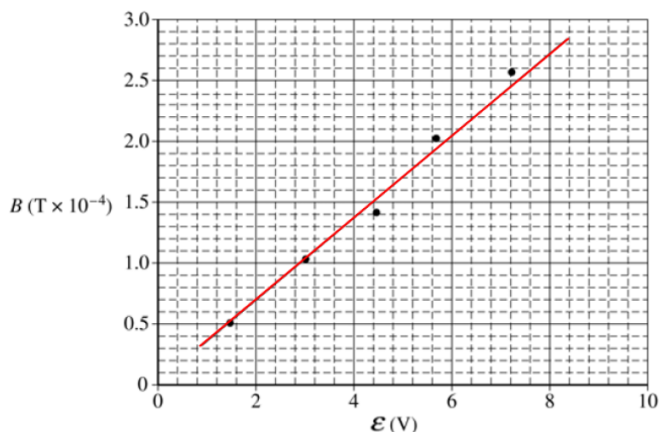
Use Ampere's law to derive an expression for the magnetic field strength at point P. Express your answer in terms of I , ℓ , N , a , and physical constants, as appropriate.

For using Ampere's law to calculate the magnetic field along the axis of the solenoid	1 point
$\int \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{enc} \therefore \left(\int \mathbf{B} \cdot d\boldsymbol{\ell} \right)_1 + \left(\int \mathbf{B} \cdot d\boldsymbol{\ell} \right)_2 + \left(\int \mathbf{B} \cdot d\boldsymbol{\ell} \right)_3 + \left(\int \mathbf{B} \cdot d\boldsymbol{\ell} \right)_4 = \mu_0 I_{enc}$	
$\left(\int \mathbf{B} \cdot d\boldsymbol{\ell} \right)_1 + 0 + \left(- \left(\int \mathbf{B} \cdot d\boldsymbol{\ell} \right)_1 \right) + Bh = \mu_0 \frac{N}{\ell} hI$	
$Bh = \mu_0 \frac{N}{\ell} hI$	
For a correct answer	1 point
$B = \frac{\mu_0 NI}{\ell}$	

Some physics students conduct an experiment to determine the resistance R_S of a solenoid with radius $a = 0.015$ m, total turns $N = 100$, and total length $\ell = 0.40$ m. The students connect the solenoid to a variable power supply. A magnetic field sensor is used to measure the magnetic field strength along the central axis at the center of the solenoid. The plot of the magnetic field strength B as a function of the emf $\mathcal{E}$ of the power supply is shown below.

- (c) LO CNV-8.C.c, SP 4.C
i. 1 point

On the graph above, draw a best-fit line for the data.



For drawing a best-fit line with at least one point above and one point below the line	1 point
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- ii. LO CNV-8.C.c, SP 6.B, 6.C
2 points

Use the straight line to determine the resistance R_S of the solenoid used in the experiment.

For calculating the slope using the best-fit line and not the data points unless they fall on the best-fit line	1 point
$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{(2.5 - 0.9)(\times 10^{-4} \text{ T})}{(6.4 - 2.0) \text{ V}} = 0.36 \times 10^{-4} = 3.6 \times 10^{-5} \text{ T/V}$	
For correctly giving the expression that relates the slope to the resistance of the solenoid	1 point
$B = \frac{\mu_0 N I}{\ell} = \frac{\mu_0 N \mathcal{E}}{\ell R} \therefore \text{slope} = \frac{\mu_0 N}{\ell R_S} \therefore R_S = \frac{\mu_0 N}{\ell \times \text{slope}}$ $R_S = \frac{(4\pi \times 10^{-7} \text{ T}\cdot\text{m/A})(100 \text{ turns})}{(0.40 \text{ m})(3.6 \times 10^{-5} \text{ T/V})} = 8.7 \Omega$	

One of the students notes that the horizontal component of the magnetic field of Earth is $2.5 \times 10^{-5} \text{ T}$.

- (d)
i. LO CNV-8.E.a, SP 2.D
1 point

Is there evidence from the graph that the horizontal orientation of the solenoid affects the measured values for B ?

____ Yes ____ No

Justify your answer.

If the line on the graph does not go through the origin, select "Yes"	
For a correct justification	1 point
Example: The horizontal component of Earth's magnetic field will add or subtract from the magnetic field of the solenoid depending on the orientation of the solenoid.	
<i>Alternate Solution</i>	<i>Alternate Points</i>
If the line on the graph does pass through the origin, select "No"	
For a correct justification	1 point
Example: Based on the graph, the line passes through the origin, so the magnetic field is zero when the emf of the power supply is zero, therefore Earth's magnetic field is not affecting the values of B .	

- ii. LO CNV-8.E.a, SP 2.E
1 point

Would the horizontal orientation of the solenoid affect the calculated value for R_S ?

___ Yes ___ No

Justify your answer.

Select "No"		
For a correct justification		1 point
Example: The horizontal component of Earth's magnetic field will not affect the change in magnetic field as the emf is changed. Therefore, the value for the resistance of the solenoid will not change.		

(e)

- i. LO FIE-6.A.b, SP 7.A, 7.C
2 points

Is the direction of the induced current in the loop clockwise or counterclockwise during the time period that the current in the solenoid is decreasing?

___ Clockwise ___ Counterclockwise

Justify your answer.

Select "Clockwise"		
For a justification indicating that the magnetic field inside the solenoid, and therefore the loop, will decrease		1 point
For a justification using Lenz's law to relate the change in magnetic field to the direction of the current		1 point
Example: As the current in the solenoid decreases, the magnetic field inside the solenoid decreases. As the solenoid's magnetic field decreases, the induced current in the loop will create a magnetic field to oppose this change. Because the solenoid's magnetic field is toward the right and decreasing, the magnetic field due to the current in the loop must be toward the right. Therefore, the current in the loop must be clockwise.		

- ii. LO FIE-6.A.d, SP 5.A, 5.E
3 points

Derive an equation for the average induced current i_{IND} in the loop during the time period that the current in the solenoid is decreasing. Express your answer in terms of I , ℓ , N , a , b , R_L , R_S , Δt , and physical constants, as appropriate.

For using Faraday's law to calculate the emf in the loop		1 point
$\mathcal{E} = \left \frac{d\Phi}{dt} \right = \frac{d(BA)}{dt} = A \frac{\Delta B}{\Delta t}$		
For using Ohm's law to calculate the current in the loop		1 point
$I_{\text{IND}} = \frac{V}{R} = \frac{A \Delta B / \Delta t}{R_L} = \frac{A \Delta B}{R_L \Delta t}$		
For using the correct radius for the area in the equation above		1 point
$I_{\text{IND}} = \frac{\pi a^2 \mu_0 N \Delta I / \ell}{R_L \Delta t} = \frac{\pi a^2 \mu_0 N I}{R_L \ell \Delta t}$		